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Transfer-learning-aided defect prediction in simply shaped CFRP specimens based on stress distribution obtained from finite element analysis and infrared stress measurement

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ABSTRACT

In this paper, we propose a framework of nondestructive testing for predicting the 3D structure of internal defects in carbon-fiber-reinforced plastic (CFRP) from the distribution of the sum of principal stresses on surfaces (DSPSS) through transfer learning. DSPSS is obtained from both the finite element method and infrared stress measurement results. Infrared stress measurements are based on Kelvin's theory to convert surface temperature changes to DSPSS changes. The machine learning model used in this framework is a 3D convolutional neural network (CNN). The transfer learning method employed in this framework is as follows. First, a CNN that predicts the 3D structure of defects is trained using the DSPSS dataset by the finite element method and the 3D structure of internal defects. DSPSS is used with noise that imitates the noise generated by experimental factors such as temperature fluctuations in infrared stress measurements and differences in physical properties between the polymer resin and the carbon fiber bundle of CFRP. Next, the CNN is trained using the DSPSS dataset obtained by infrared stress measurement and the 3D structure of defects. The accuracy of the trained CNN is evaluated using DSPSS infrared stress measurements. We discuss the factors that enable us to predict the 3D defect data from the two-dimensional DSPSS using a variational autoencoder. The proposed method makes it possible to estimate internal defect information.

1. Introduction

Carbon-fiber-reinforced plastic (CFRP) is a composite material consisting of a resin matrix and carbon fiber reinforcement. CFRP is widely used in the aerospace industry because of its high rigidity, high strength, and lightweight. For example, more than 50% of the fuselage structure of the Boeing 787 is made of CFRP [1]. The material is also commonly used in laminates of prepreg, a unidirectional reinforcing material with high strength and stiffness in one direction [2].

Defects of laminates are extremely complex, including delamination, fiber fracture, and base matrix cracking, requiring highly efficient and accurate defect detection. In general, nondestructive testing can be performed by methods such as ultrasonic measurement [3–8] and radiation transmission [9–13] methods. They require a great deal of labor, time, safety control, economic burden, and experience of engineers in measurements. In addition to the existing nondestructive testing methods described above, infrared thermography is also used [14–24].

Infrared thermography is a method of measuring the infrared energy emitted from the surface of an object using an infrared sensor. The measured infrared energy is converted into a temperature distribution. Compared with existing nondestructive analysis methods, infrared thermography has the following advantages: no safety control is required, analysis and time costs are low, no contact medium is needed, and the skill of the test engineer does not affect the results.

In recent years, the distribution of the sum of principal stresses on surfaces (DSPSS) is often calculated from temperature changes under cyclic loading on the basis of the thermoelastic effect. The change in temperature is obtained by infrared analysis based on the thermoelastic effect. The infrared stress measurement method is based on Kelvin's theory to calculate DSPSS from the surface temperature changes generated by repeatedly applying a load to an object. When an object is

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in an adiabatic state in the elastic region, the amount of change in DSPSS and the amount of change in temperature are proportional. The advantages of nondestructive testing using an infrared camera include the following: there is no need for safety management, measurement is possible even when the analysis object is large, and the skill of the test engineer does not affect the results. These advantages allow the analysis to be performed while eliminating the shortcomings of existing nondestructive testing. This method is the infrared stress measurement.

In some studies, infrared stress measurement is used to search for defects. Sakagami et al. confirmed that the infrared stress measurement of structures is useful for detecting defects in bridges [25]. Jang et al. constructed a convolutional neural network (CNN) model by transfer learning to predict concrete cracks from hybrid images. They combined vision and infrared thermography images to achieve high accuracy [26]. De Finis et al. built a new experimental model for evaluating and predicting material stiffness degradation by simply using thermoelastic data of quasi-isotropic CFRP [27]. Zhang et al. proposed a novel method for defect detection in thermographic inspections using the Differentiated Tensor Low Rank Soft Decomposition approach. By extracting distinct low-rank information from thermal data, their technique significantly improves defect visibility and detection accuracy in composite materials, including CFRP, even in the presence of noise and optical interference [28]. Xue et al. developed an innovative approach to non-destructive evaluation of thermography data using a few-shot learning segmentation model. By employing a memory-linked domain transfer strategy, their method enhances defect detection in CFRP composites with limited training data [29].

To predict the information of defects of a CFRP structure, machine learning is used in some studies. Bellouard et al. divided a CFRP laminate into 10 parts in the longitudinal direction and conducted numerical analysis and experiments to obtain natural frequencies. They then developed a neural network model to predict the location of defects [30]. As a result, a machine learning model was developed to determine the two-dimensional location and amount of defects using first- to third-order natural frequencies as input data. Saeed et al. proposed a framework that can automatically detect defects from given thermograms [31]. Wang et al. proposed a CNN-based image semantic segmentation technique for pixel-level defect classification using digital image correlation strain field images [32]. Hasebe et al. predicted crash-induced damage from the surface properties of CFRP laminates with experimental damage using decision-based multitask learning and compared the accuracy of their method with those of existing methods [33].

We have conducted a proof-of-concept study to predict the 3D structure of a defect from DSPSS using the finite element method (FEM) data [34]. We predicted the 3D defect structure of a simply shaped CFRP specimen with a teflon sheet inserted between the prepreg layers as a defect, using a CNN [35]. However, the defect information was not predicted using the experimentally obtained DSPSS but the FEM-based DSPSS. Although FEM is a powerful analysis tool for revealing a real phenomenon, it is necessary to predict defects using experimentally obtained data for practical applications. Considering practical applications, a machine learning model that is robust for the measurement of gaps and errors is needed. Therefore, a combination of FEM and experimental results is necessary for the training of machine learning.

In this study, in order to combine the results of experiments and FEM for the training of a machine learning model, a CNN to predict defect information is trained in two steps. Firstly, a CNN is trained using FEM-based DSPSS and datasets of defect information, for which a large number of data can be prepared. Secondly, the transfer learning [36] of the CNN prepared in step 1 is conducted using the dataset of DSPSS and defect information from the experiment, in which a small number of data are available. Before training using the experimental dataset, the CNN is trained using the FEM dataset. From the prediction of the defect information by the trained CNN, we show that using the parameters obtained from the FEM dataset makes it possible to obtain a highly versatile CNN for the experimental dataset.

2. Theory

2.1. Infrared stress measurement

In a solid, a temperature change is measured according to the stress change. This phenomenon is called the thermoelastic effect. The relationship between the temperature change ΔT and the DSPSS change $\Delta\sigma$ under cyclic loading is shown by following equation:

$$\Delta T = -\frac{\alpha}{\rho C_p} T \Delta\sigma, \tag{1}$$

where α is the coefficient of thermal expansion, ρ is the mass density, C_p is the specific heat at constant pressure, and T is the absolute temperature. Eq. (1) is known as Kelvin's thermoelastic theory. Using this theory, we obtain DSPSS by infrared stress measurement [37].

2.2. CNN

A CNN [38] is a neural network with a convolutional layer in the model. The convolutional layer is a function that maps the input tensor $\boldsymbol{x}$, which consists of two axes (height, width), to the output tensor $\boldsymbol{y}$, which also consists of two axes (height, width), using convolution operations. The mathematical expression is shown in the following equation:

$$y_{ij} = \sum_s \sum_t W_{st} x_{(i+s)(j+t)} + b, \tag{2}$$

where $\boldsymbol{W}$ is a kernel and represents the height, width, output channel, and input channel in that order on each axis, and $\boldsymbol{b}$ is the bias term. The model parameters of the convolution layer are $\boldsymbol{W}$ and $\boldsymbol{b}$.

The CNN is a specialized neural network used to process data with a grid structure. Examples of data with a lattice structure include time-series image data consisting of one-dimensional arrays of samples acquired at equal time intervals and image data consisting of two-dimensional arrays of pixels.

A typical convolutional network layer consists of three steps. The first step performs multiple convolutional operations in parallel and outputs a set of linear activation values. In the next step, each linearly activated value is processed nonlinearly using an activation function. In the last step, the output is modified using a pooling function.

2.3. Transfer learning

The features acquired by CNNs trained on large datasets such as ImageNet [39] are very versatile and have been studied for applications in not only the same domain but also other domains by conducting transfer learning [36,40–45]. One of the advantages of transfer learning is that features learned using a large number of high-quality data can be applied to the other domains with a small number of data. In the following, we describe the transfer learning methods utilized in this study.

Fine-tuning is one of the transfer learning concepts. In deep learning, the first few layers are used to identify task features. In transfer learning, the output layer of the trained model is removed and a new layer is added to adapt it to the new domain. This means that fine-tuning requires slightly more learning, but it is faster and more accurate than building a model from scratch.

3. Method

3.1. Flow of inverse estimation method

The flow of the proposed inverse defect estimation method is shown in Fig. 1. The method is based on a CNN trained using the DSPSS obtained from FEM. Firstly, DSPSS is calculated by FEM. Next, a distribution of noise imitating the infrared stress measurement is added. A CNN is then trained to predict the 3D information of the defect using

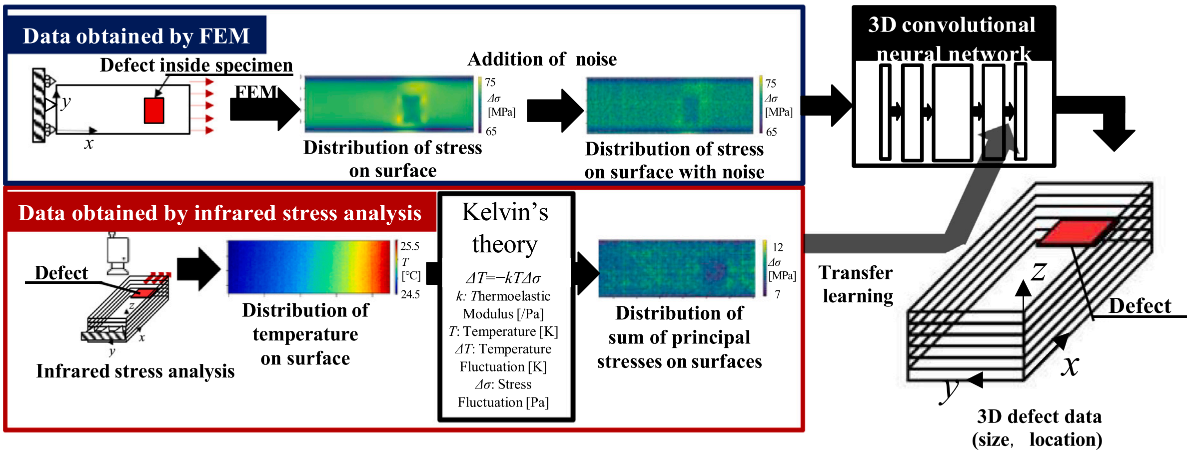

Fig. 1. Flow of proposed inverse estimation method.

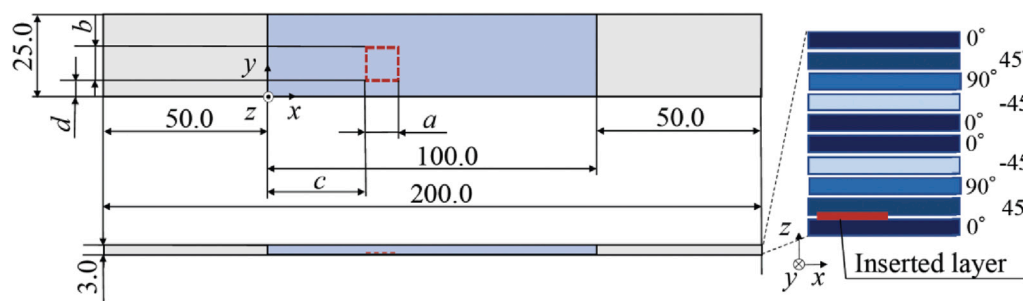

Fig. 2. Example of CFRP specimen with model defect.

Table 1
Prepared CFRP specimens in infrared stress measurement.

	Without defect	With defect											
Inserted between	–	9–10	6–7	8–9	7–8	8–9	6–7	9–10	7–8	7–8	6–7	7–8	6-7
Size (a mm × b mm)	–	3 × 3		3 × 5		5 × 10		10 × 10		10 × 15		15 × 15	
x position (c mm)	–	40	20	80	40	10	40	60	40	70	80	40	20
y position (d mm)	–	10	2.5	5	7.5	2.5	6	5	2.5	2.5	7.5	7.5	2.5
Number of specimens	17	1	1	1	1	1	1	1	1	1	1	1	1

Table 2
Fatigue testing conditions.

Load condition	Repetitive load
Maximum load	2.0 kN
Minimum load	0.2 kN
Wave form	Sine wave
Frequency	5 Hz

the DSPSS of FEM with added noise. The trained CNN is further trained with the DSPSS obtained by infrared stress measurement. Finally, the transfer learning of the CNN is performed using the DSPSS obtained by infrared stress measurement. The pretraining of the CNN with the DSPSS of FEM covers the DSPSS of infrared stress measurements, for which only a small number of data sets are available. In this study, only the output layer model parameters are retrained.

3.2. Infrared stress measurement

In this study, the infrared stress measurement was used to obtain DSPSS for the training of a machine learning model. An example of a simply shaped CFRP specimen used in the infrared stress measurement is shown in Fig. 2. The specimen consisted of 10 layers of unidirectionally reinforced composites. The fiber directions of the unidirectional reinforced composites that were laminated were 0°, 45°, 90°, −45°, 0°, 0°, −45°, 90°, 45°, and 0°. The matrix was epoxy. The dimensions of the laminate were 200.0 mm × 25.0 mm × 3.0 mm, and the defect was inserted in the rectangle with a red dashed line. We prepared the specimens shown in Table 1 by changing the defect length a, the width b, the insertion position in the x-direction c, the insertion position in the y-direction d, and the insertion layer. Glass-fiber-reinforced plastic tabs were attached at the gray area in Fig. 2 so that the specimen could be inserted into the jig of the fatigue testing apparatus. 17 specimens were without any defect and 12 specimens had a defect. The surface of each specimen was painted black to prevent measurement distortions caused by reflections. A Teflon sheet was inserted into specimens with a defect.

The infrared stress measurement setup used for the experiments is shown in Fig. 3. The fatigue testing apparatus used was the 810 Material Test System. The infrared camera was an FLIR SC7500 thermal imaging camera. The fatigue testing apparatus was set up using the conditions shown in Table 2. A tensile–tensile load was repeatedly applied with a maximum load of 2.0 kN and a minimum load of 0.2 kN. The load waveform was a sine wave and the frequency was 5 Hz. The resolution of the infrared thermography camera was 320 × 256. The number of frames used for the lock-in process was 4000.

The resolution of thermal images was 150 × 50. Standardized images were used for obtaining the DSPSS for machine learning. DSPSS was measured on one side of CFRP specimens as shown in Fig. 4. The obtained DSPSS is shown in Fig. 5.

Table 3
Property values for each region. Young's modulus E and shear modulus G are expressed in MPa. The values are set on the basis of the physical properties of the materials used in the experiments.

	E_1 [GPa]	E_2 [GPa]	E_3 [GPa]	ν_{12}	ν_{13}	ν_{23}	G_{12} [GPa]	G_{13} [GPa]	G_{23} [GPa]
CFRP	136.6	9.65	9.65	0.29	0.29	0.40	5.2	5.2	3.4
Defect	200.0	200.0	200.0	0.39	0.39	0.39	71.9	71.9	71.9

Table 4
FEM analysis conditions.

Load condition	2.0 kN along x-axis
Tensile conditions	Load along x-axis
Binding condition	yz plane constraint

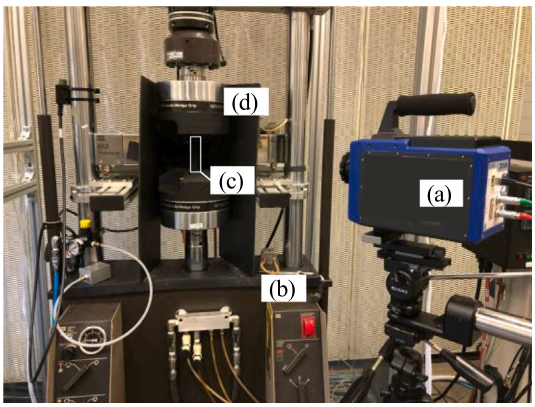

Fig. 3. Equipment used in experiment. The system consists of (a) an infrared camera, (b) a fatigue testing apparatus, (c) a CFRP specimen, and (d) a blackout curtain to prevent reflections.

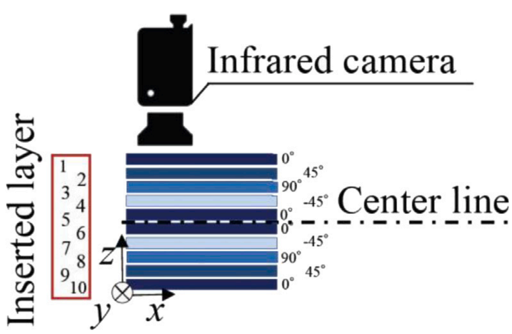

Fig. 4. Schematic of interlayer position of specimen and position of infrared camera.

3.3. FEM analysis

In this study, FEM was used to obtain the DSPSS for training in machine learning. The properties of the CFRP and defect are shown in Table 3. The defect representation method is shown in Fig. 6. Double nodes were assigned to black dots to reproduce the experimental setup for infrared stress measurements where shear does not propagate. The physical properties of the defect were based on the properties of Teflon. The mesh used in the analysis was orthogonally equidistant with dimensions of 0.5 mm × 0.5 mm × 0.1 mm. The total number of nodes was 317,781. The analysis conditions are shown in Table 4 and Fig. 7. We set the yz plane constraint and added a 1.0 kN load along the x-axis.

The DSPSS obtained by FEM under the aforementioned defect representation method was compared with that obtained by infrared stress measurement. In this study, as shown in Fig. 8, the area enclosed by the red square through the defect was trimmed in step 1. In step 2, the stress values were averaged, profiled in the y-direction, and compared with the DSPSS obtained by both infrared stress measurement and FEM analysis. This DSPSS was standardized in the same way as that obtained by infrared stress measurement.

3.4. Dataset used in machine learning

By infrared stress measurement and FEM analysis under the conditions in 3.2 and 3.3, we obtained a dataset that pairs the 3D information of a defect with DSPSS. The pixel data shape of the defect is 150 × 50 × 5, and that of DSPSS is 150 × 50, whose real size is 75.0 mm × 25.0 mm. By changing the position and size of the defects in the defect insertion area shown in Fig. 9, we have prepared one DSPSS without defects and 1032 DSPSSs with defects as shown in Table 5. The thermal noise generated in infrared stress measurements and the noise generated by differences in the physical properties of the resin and carbon fiber bundle are added to the stress distributions by FEM analysis. To the DSPSS without any defect, 5000 types of noise are added, and to the DSPSS with a defect, 5 types of noise are added. As a result, 5000 DSPSSs without a defect and 5165 DSPSSs with a defect are produced.

The training and test data of the DSPSS obtained by infrared stress measurement are shown in Fig. 10. The datasets are assigned without any bias in the defect insertion layer.

3.5. Training for CNN

The information on three-dimensional defects is represented as shown in Fig. 11. When a new layer is added, 0 is placed if there is a defect in the stacking direction, and 1 if there is no defect. This process ensures that the sum of the values in the stacking direction is 1. Therefore, this problem is solved as a classification problem in the depth direction. In this study, the three-dimensional defect information composed of 0 and 1, as shown in Fig. 11, represents the predicted defect location, size, and shape. This process allows the sum of the values in the stacking direction to be 1. Thus, the problem is solved as a classification problem in the depth direction. In addition, normalization is performed on a data-by-data basis. In this study, a CNN trained with the noisy DSPSS obtained by FEM is subjected to transfer learning using the DSPSS obtained by infrared stress measurement. The architecture and process of CNN training using the noisy DSPSS obtained by FEM are shown in Fig. 12(a). The training conditions of the CNN are shown in Table 6. The training data consist of 3000 datasets without any defect and 3095 datasets with a defect. The validation and test data consist of 1000 datasets without any defect and 1035 datasets with a defect. 64 images composed one minibatch, and the loss function is evaluated and weights are updated 800 times. The training is performed by repeating the evaluation of the loss function and updating the weights 800 times.

We perform the transfer learning of the CNN using the DSPSS obtained by infrared stress measurement; the architecture and training process of the CNN are shown in Fig. 12(b). The training conditions of transfer learning of the CNN are shown in Table 7. The training data consist of 10 datasets without any defect and 12 datasets with a defect. Test data consist of 6 datasets without any defect and 6 datasets with a defect. One image constituted one minibatch, and the training is performed by repeating the loss function evaluation and updating the weights 1000 times.

Table 5
Prepared CFRP specimens for FEM.

	Without defect	With defect					
Inserted between	–	Any of the layers between the 6th and 10th layers					
Size (a mm × b mm)	–	3 × 3	3 × 5	5 × 10	10 × 10	10 × 15	15 × 15
Number of specimens	1	480	288	96	96	48	24

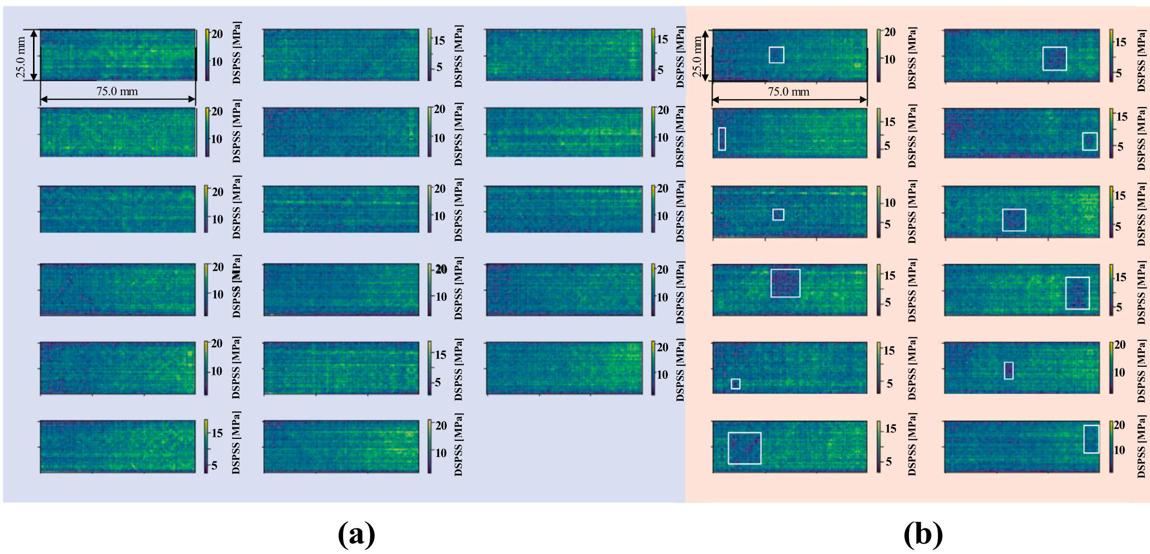

Fig. 5. DSPSS obtained by infrared stress measurement. (a) Stress distribution of without defect specimen measured from camera position 1 in Fig. 4 and (b) stress distribution of with defect specimen measured from camera position in Fig. 4. The white dashed line is the area where the defect is inserted.

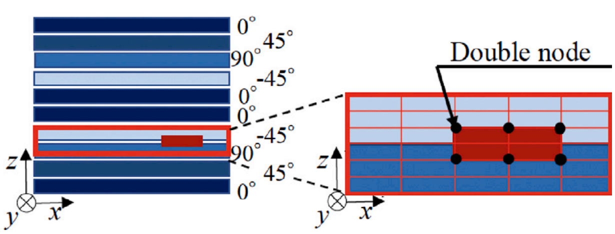

Fig. 6. Defect representation method employed in this study. Double nodes are assigned to black dots to reproduce the experimental setup for infrared stress measurements where shear does not propagate.

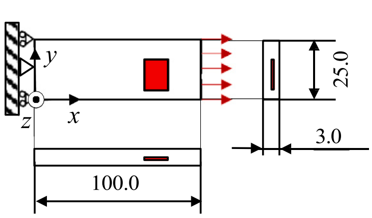

Fig. 7. FEM analysis conditions. The red area shows the inserted defect. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.6. Evaluation of prediction results from transfer learning

Two quantitative evaluation indices are used to evaluate the prediction of the 3D information of a defect by transfer learning. One is the percentage of agreement in the prediction of the defect plane position R and the other is that in the prediction of the defect insertion layer P.

The consistency of the prediction of the defect planar position R is determined by calculating the percentage of results predicting that defects are in the defect region of the ground truth.

$$R = \frac{N_{\text{black}}}{N_{\text{red}}} \tag{3}$$

Here, N_{red} is the number of pixels in the defect area as shown in Fig. 13. N_{black} is the number of pixels predicted as a defect in the area extending 2 mm from the red frame as shown in Fig. 13. The reason for the 2.0 mm extension from the red frame is to take into account the coordinate misalignment between the original coordinates and DSPSS obtained by infrared stress measurement when image cropping and compression are conducted. A schematic of method of the calculating the probability of the existence of a layer with a defect is shown in Fig. 14. The probability of the presence of a defect in each layer is calculated by dividing the total number of pixels predicted as a defect in each layer by the number of pixels predicted as a defect in all layers.

3.7. Evaluation of features of dataset

The 3D structure of a defect is accurately predicted from the DSPSS, which is a two-dimensional data. Here, we consider the features of datasets by observing the latent variable space, which is a low-dimensional representation of the characteristics of the defect information and DSPSS, using a variational autoencoder (VAE). The VAE is a generative model that takes the form of data sampled from a latent variable that follows a standard normal distribution. When clustering by defect insertion layers is possible, the dataset used in this study can be predicted the defect insertion layer while small defects close to the DSPSS are measured and large defects far from the surface provide a similar DSPSS. In the following, we construct a VAE on the DSPSS and defect information to qualitatively analyze the features of

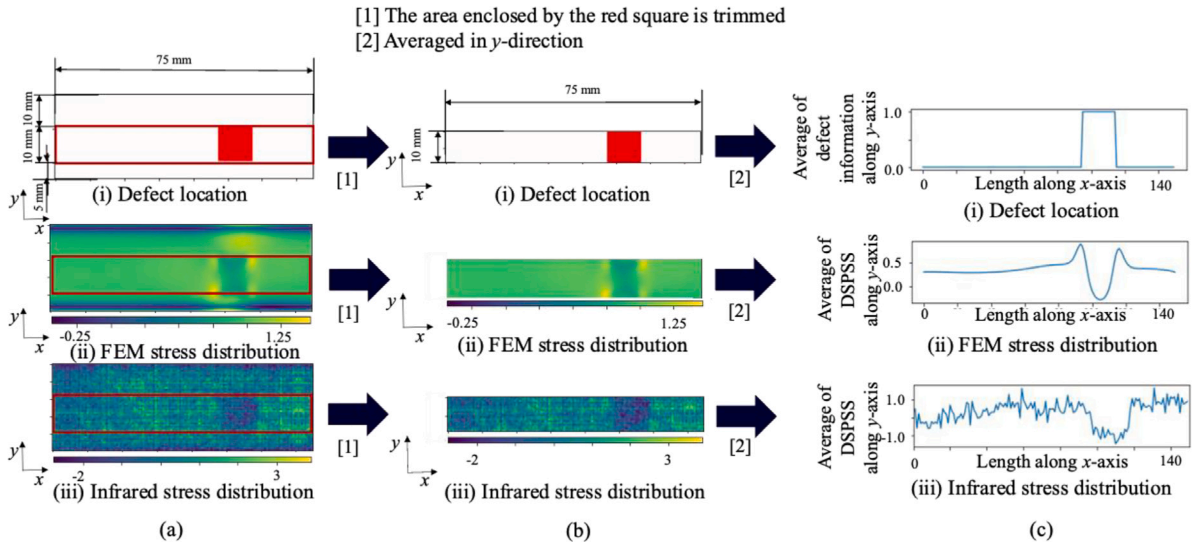

Fig. 8. Method of comparing DSPSS obtained by infrared stress measurement with that obtained by FEM. (a) Original distributions of (i) defect location, (ii) stress distribution obtained by FEM and (iii) stress distribution obtained by infrared stress measurements. (b) Trimmed distribution of (i) defect location, (ii) stress distribution obtained by FEM and (iii) stress distribution obtained by infrared stress measurements. (c) Profiled results of (i) defect location, (ii) stress distribution obtained by FEM and (iii) stress distribution obtained by infrared stress measurements. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

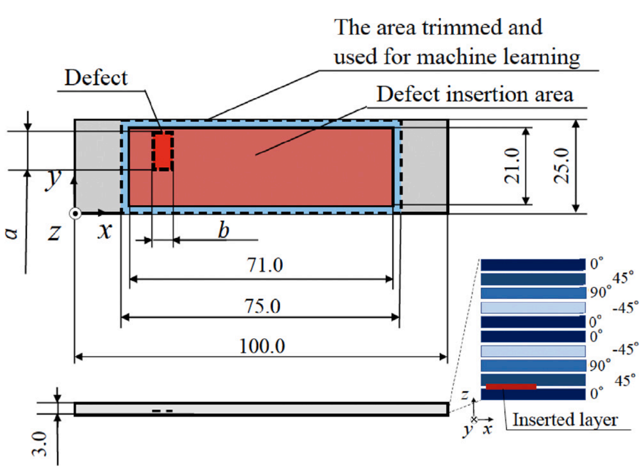

Fig. 9. Example of CRRP specimen with defect for FEM.

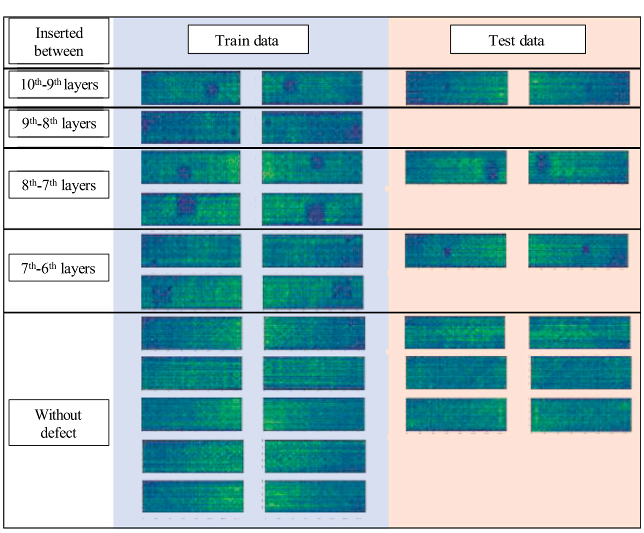

Fig. 10. Training and test data for DSPSS obtained by infrared stress measurements. Data augmentation with 180° rotation.

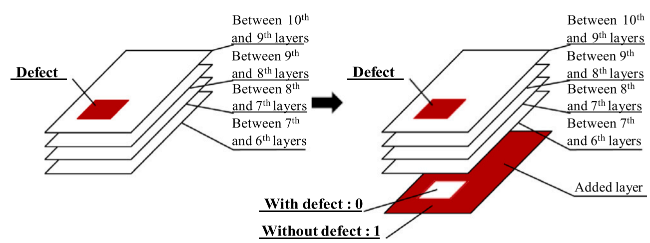

Fig. 11. Example of defect data used in inverse analysis.

Table 6
Training conditions for CNN using noisy DSPSS obtained by FEM.

Loss function	Binary cross-entropy
Optimizer	Adam (Lr 0.00007)
Batch size	64
Number of training epochs	800
Number of training data	Without defect: 3000 With defect: 3095
Number of validation data	Without defect: 1000 With defect: 1035
Number of test data	Without defect: 1000 With defect: 1035

the datasets, and we visualize and discuss the latent variable space. The FEM DSPSS data are used for the training of the VAE because they can be prepared in large numbers and easily clustered in a latent variable space.

3.7.1. VAE using DSPSS

The VAE using the DSPSS consists of an encoder for the low-dimensional representation of the DSPSS and a decoder for reproducing the DSPSS from the low-dimensional representation of latent variables. A schematic of the VAE using the DSPSS is shown in Fig. 15(a). Table 8 shows the training conditions for the VAE.

3.7.2. VAE using defect information

The VAE using the defect information consists of an encoder and a decoder. The encoder reproduces the low-dimensional representation

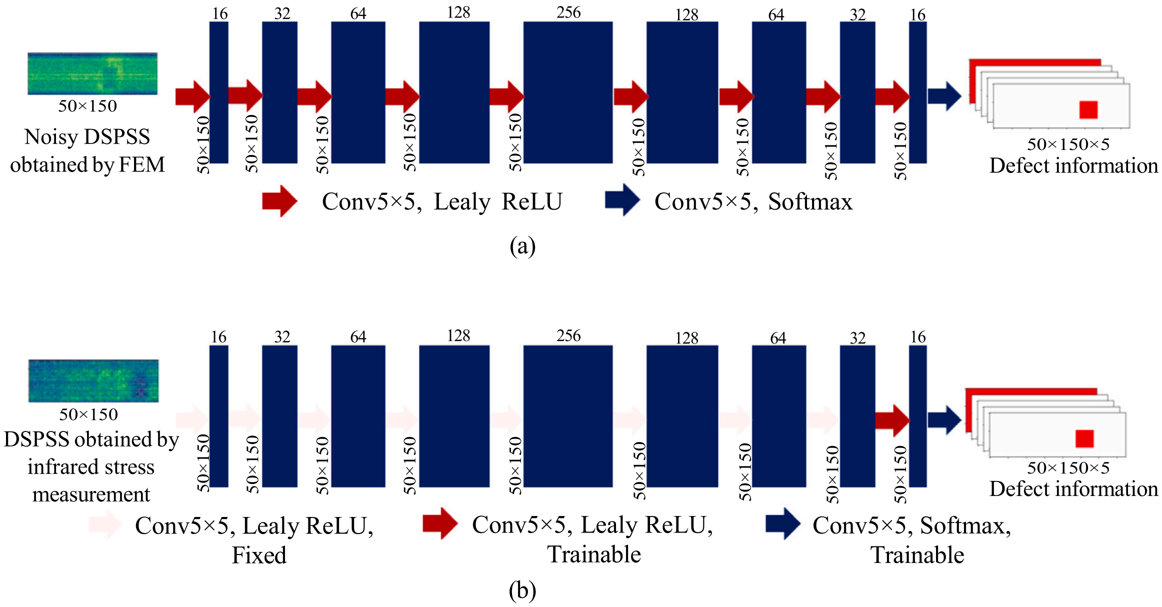

Fig. 12. Architecture of CNN. (a) Training process using noisy DSPSS obtained by FEM and (b) transfer learning and training process using DSPSS obtained by infrared stress measurement.

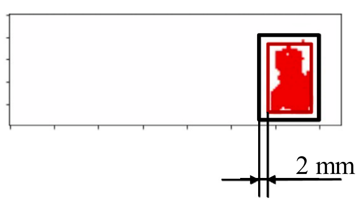

Fig. 13. Schematic of coincidence rate R of prediction of defect plane position. The rectangle with the red line is the defect area and the rectangle with the black line is the area extending 2 mm from the defect area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7
Training conditions for CNN using DSPSS obtained by infrared stress measurements in transfer learning.

Loss function	Binary cross-entropy
Optimizer	Adam (Lr : 0.00007)
Batch size	1
Number of training epochs	1000
Number of training data	Without defect: 10 With defect: 6
Number of test data	Without defect: 6 With defect: 6

Table 8
Training conditions for VAE using DSPSS.

Objective function	L_{ELBO}
Optimizer	Adam (Learning rate : 0.0007)
Batch size	16
Number of training epochs	1000
Number of training data	FEM stress distribution of defect size observed by infrared stress measurements: 206
Dimensions of latent variable space	1000

of the defect information and the decoder reproduces the defect information from the low-dimensional representation of latent variables. A schematic of the VAE using the defect information is shown in Fig. 15(b). Table 9 shows the training conditions for the VAE.

Table 9
Training conditions for VAE using defect information.

Objective function	L_{ELBO}
Optimizer	Adam (Learning rate : 0.0007)
Batch size	16
Number of training epochs	1000
Number of training data	3D information of defect size observed by the infrared stress measurements: 206
Latent variable space dimension	3000

4. Results and discussion

In this section, we first present the results of the comparison between the infrared stress measurement and the FEM analysis using the profile method as shown in Fig. 10. Next, we present the prediction results of a CNN that predicts the 3D information of a defect from stress distributions with noise obtained by FEM. Finally, we summarize the prediction results of a CNN with transfer learning using the DSPSS obtained by infrared stress measurement using the VAE.

4.1. Comparison of DSPSS obtained by infrared stress measurement and that by FEM

The results of the comparison between the infrared stress measurement and the FEM analysis using the above profile method are shown in Fig. 16. A decrease in stress at the defect insertion area is observed in both the FEM and infrared stress measurement data. Furthermore, it is confirmed that the stress profiles are consistent for all data regardless of the defect insertion layer. Therefore, using this defect representation method, DSPSS is prepared by FEM analysis for a model interpolating the experimentally prepared specimen.

4.2. Prediction of 3D information of defect from stress distribution with noise obtained by FEM (Test data)

In this section, we visualize and compare the output (predicted data) and the teacher data (ground truth) when predicting the 3D information of a defect from the stress distribution with noise obtained by FEM. Fig. 17 shows the accuracy and binary cross-entropy loss plot

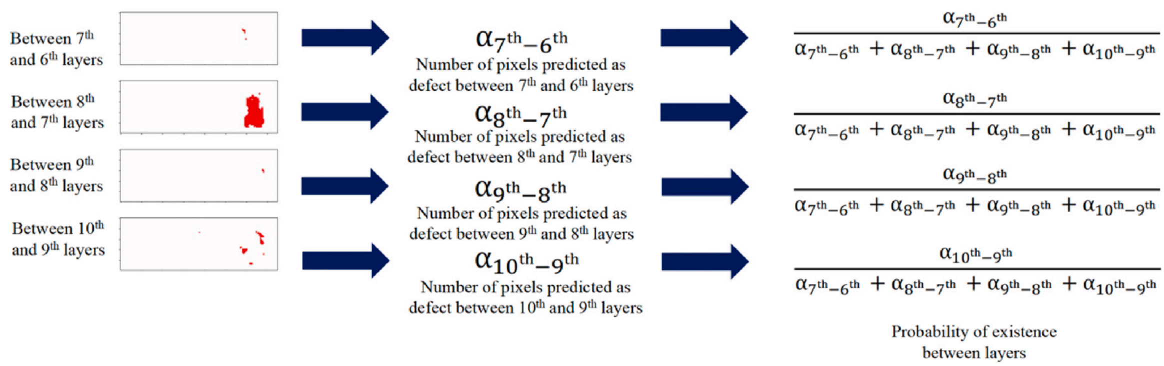

Fig. 14. Schematic of predicted defect insertion layer P.

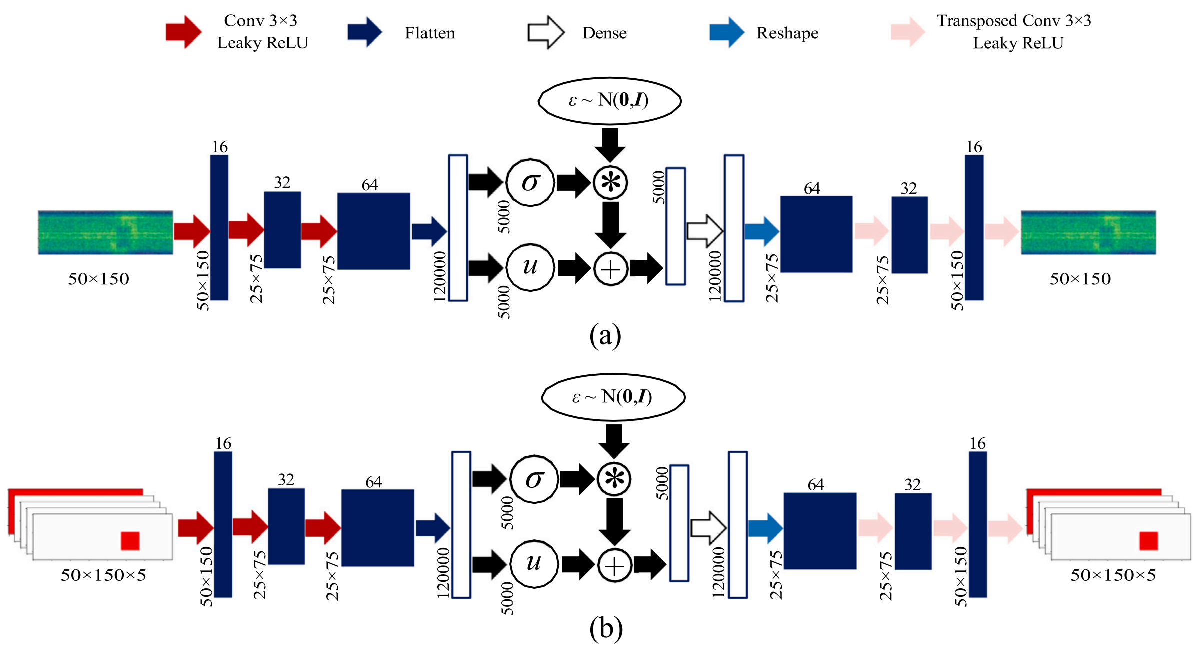

Fig. 15. Schematic of VAE using (a) DSPSS and (b) defect information. σ is the standard deviation, u is the mean, ϵ is the value following a normal distribution N, and $*$ and $+$ are symbols for the product and sum, respectively.

during pre-training of the CNN model using FEM stress distribution with noise. It can be confirmed that both train loss and validation loss decrease properly. In this study, pre-training of the CNN model was completed before the loss fully converged to improve performance during transfer learning. Fig. 18 shows the stress distribution obtained by FEM, the predicted data of the CNN, and the ground truth for the test data with and without a defect inserted. It can be seen that for each defect size, the plane position of the defect and the inserted layer are correctly predicted. These results indicate that the 3D information of the defect is accurately predicted for the FEM DSPSS with the addition of pseudo-noise generated by infrared stress measurement. Therefore, we will use this CNN model parameter for transfer learning.

4.3. Transfer learning using DSPSS obtained by infrared stress measurement

Transfer learning is performed on a CNN trained with the noisy FEM DSPSS, using the DSPSS obtained by infrared stress measurement. Fig. 19 shows the accuracy and binary cross-entropy loss plot during transfer learning of the CNN model using DSPSS obtained from infrared stress measurement. It can be confirmed that both train loss and validation loss decrease properly. The training was terminated before the validation loss increased. Due to the extremely small number of experimental data, the same data were used for both validation and test data in this study. In transfer learning, as shown in Fig. 12, we retrain only the model parameters of the output layer side, allowing the training of the CNN model to be supplemented with FEM data despite the limited number of experimental data. Ultimately, when predicting defect information using DSPSS obtained from infrared stress measurements, all model parameters are used for the prediction.

We visualize and compare the output (predicted data) with the teacher data (ground truth) for the test data not used for training. Fig. 20 shows an example of the DSPSS obtained by infrared stress measurement, the predicted data from the CNN, and the ground truth for test data with and without a defect inserted.

These results indicate that transfer learning using the DSPSS obtained by infrared stress measurement can accurately determine specimens with and without a defect. Furthermore, for the specimens with a defect, the inserted layer can be approximately predicted. It can also be seen that the specimens with a defect are predicted to have the defect in several layers owing to the noise in the DSPSS obtained by infrared stress measurement and changes in stress at the contours. On the other hand, the prediction of the plane position of the defect is approximately correct. For this reason, we quantitatively evaluated the prediction results for specimens with a defect using both the agreement ratio of the plane position predictions R and the predicted P of the defect insertion layer. The predicted R and predicted P for each test data are shown in

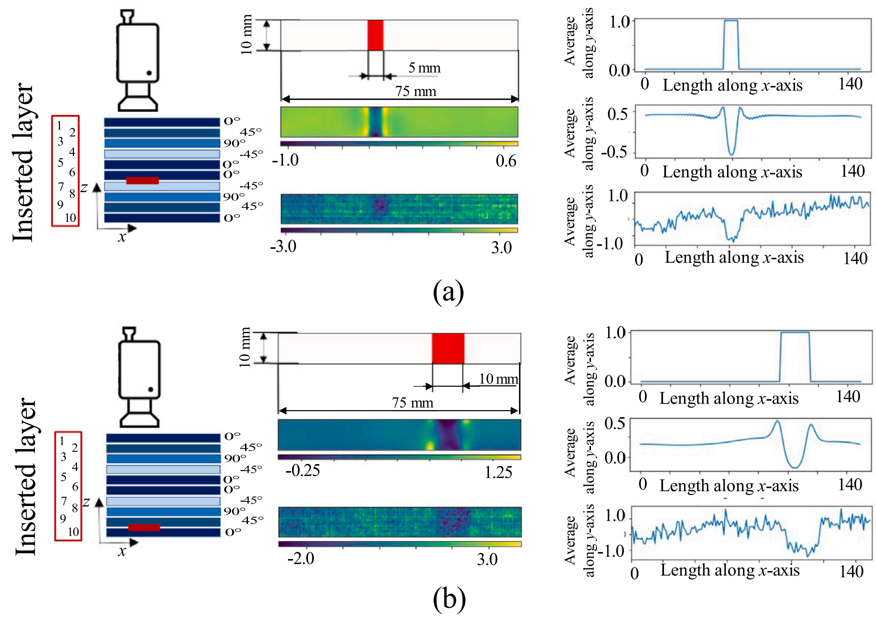

Fig. 16. Comparison of DSPSS obtained by infrared stress measurement and that obtained by FEM. (a) With defect inserted between layers 7 and 6 and (b) with a defect inserted between layers 10 and 9.

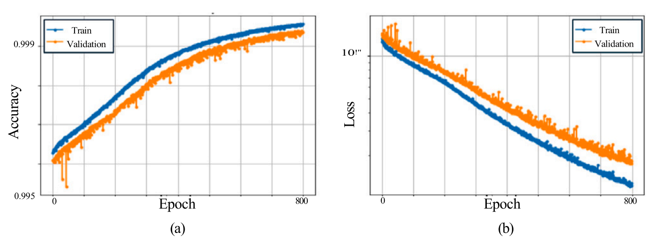

Fig. 17. Plot of (a) accuracy and (b) loss for training using DSPSSs with noise obtained by FEM.

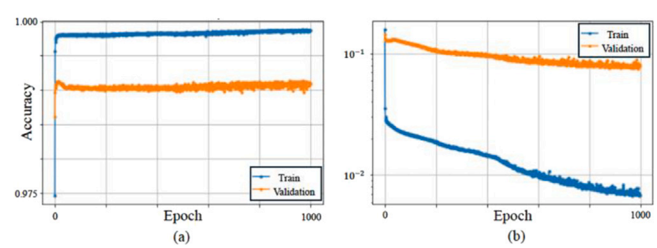

Fig. 19. Plot of (a) accuracy and (b) loss for transfer learning using DSPSS obtained by infrared stress measurement.

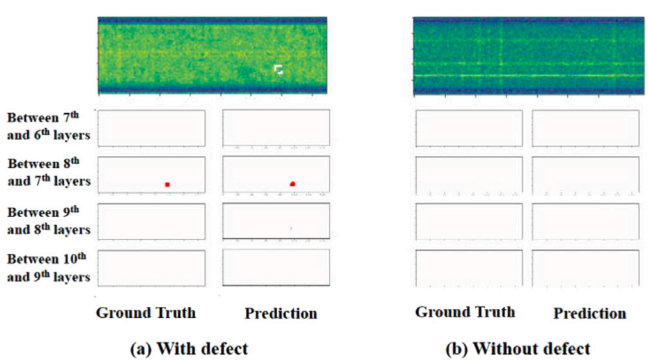

Fig. 18. (a) Stress distribution obtained by FEM (top), ground truth (bottom left), and predicted data (bottom right) for test data with defect and (b) stress distribution obtained by FEM (top), ground truth (bottom left), and predicted data (bottom right) for test data without defect. The white dashed line is the area where the defect is inserted.

Table 10. It is confirmed that the predicted R is within 2 mm wider than the ground truth, although it is numerically inferior when the size of the defect is small. Furthermore, when the position of the defect insertion layer is calculated from the established position of the defect insertion layer, the results showed that the P in the correct layer is greater than 0.5 for the test data with all the defects inserted. We discuss the factors that enabled the prediction of defect information, which is 3D data, from the DSPSS, which is two-dimensional data.

4.4. Evaluation of features of dataset using VAE

The plots of KL loss, reconstruction loss and ELBO loss during the training of the VAE for DSPSS and defect information are shown in Figs. 21(a) and 22(a), respectively. In both cases, it can be confirmed that the ELBO loss has sufficiently decreased. The plots of reconstruction accuracy during the training of the VAE for DSPSS and defect information are shown in Figs. 21(b) and 22(b), respectively. In both cases, the reconstruction is performed with sufficient accuracy. In this study, we used a VAE to examine the structure of the dataset, and therefore, we prepared only the training data.

4.4.1. Evaluation of latent variable space of DSPSS using VAE

The dimension of the latent variables required to reproduce the DSPSS using the VAE is 1000. Here, the latent variable space is visualized in a two-dimensional space using a dimensional compression technique called UMAP. The result of the visualization, color-coded by the layer number where defects exist, defect width, and defect height, is shown in Fig. 23. The figure shows that the data are independent and can be classified by the layer in which the defect is inserted and the width and height of the defect.

Fig. 23(a–i) shows that all the characteristics of the DSPSS can be determined according to the layer number where the defect is present. Defects between layers 10 and 9 and between layers 7 and 6 are in the same neighborhood in Fig. 23(a–i), but it is possible to distinguish them using another VAE as shown in Fig. 23(a–ii). As the characteristics of the DSPSS vary depending on the defect insertion layer, a large defect

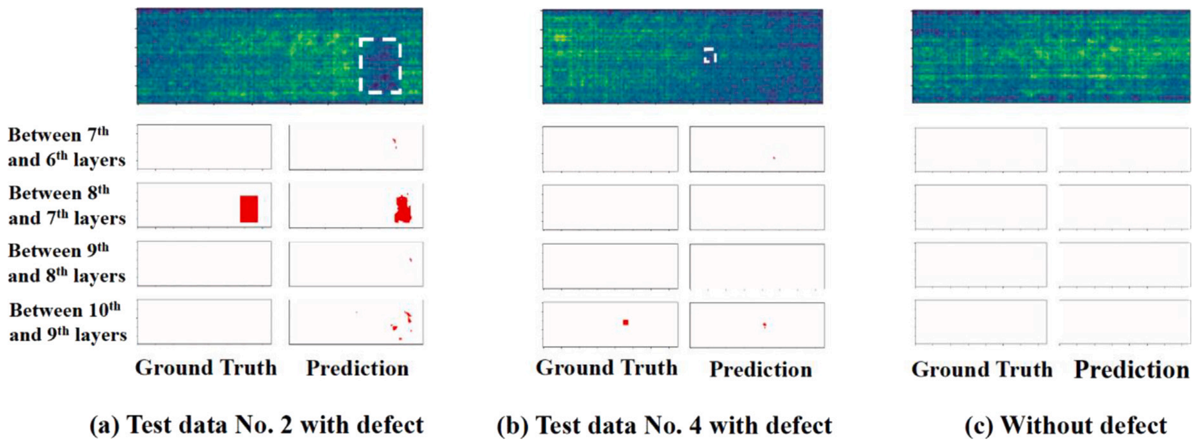

Fig. 20. (a), (b) DSPSS (top), ground truth (bottom left), and predicted data (bottom right) obtained by infrared stress measurements for test data with defect inserted and (c) DSPSS (top), ground truth (bottom left), and predicted data (bottom right) obtained by the infrared stress measurements for test data without defect inserted. The white dashed line is the area where the defect is inserted.

Table 10
Agreement ratio of plane position prediction R and prediction of defect insertion layer P.

		Test Data No. 1		Test Data No. 2		Test Data No. 3		Test Data No. 4		Test Data No. 5		Test Data No. 6	
		Prediction	Ground Truth	Prediction	Ground Truth	Prediction	Ground Truth	Prediction	Ground Truth	Prediction	Ground Truth	Prediction	Ground Truth
R		0.56	1.00	0.83	1.00	0.61	1.00	0.31	1.00	0.82	1.00	0.59	1.00
P	7-6 layers	0.21	0.00	0.02	0.00	0.56	1.00	0.21	0.00	0.01	0.00	0.52	1.00
	8-7 layers	0.00	0.00	0.84	1.00	0.00	0.00	0.00	0.00	0.70	1.00	0.08	0.00
	9-8 layers	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00
	10-9 layers	0.79	1.00	0.13	0.00	0.44	0.00	0.79	1.00	0.27	0.00	0.41	0.00

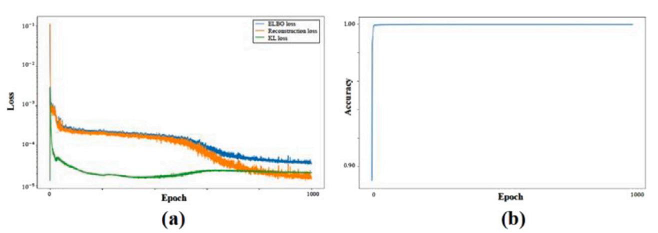

Fig. 21. (a) KL loss, reconstruction loss and ELBO loss during the training of the VAE for DSPSS and (b) KL loss, reconstruction loss and ELBO loss during the training of the VAE for DSPSS.

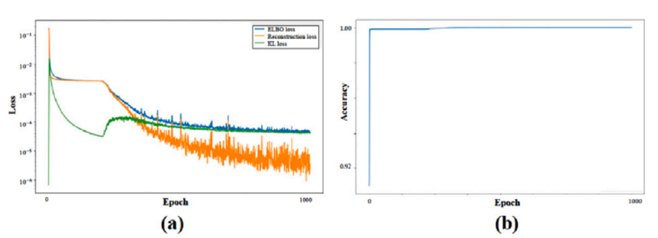

Fig. 22. (a) KL loss, reconstruction loss and ELBO loss during the training of the VAE for defect and (b) KL loss, reconstruction loss and ELBO loss during the training of the VAE for defect.

is located far from the infrared camera and a small defect is located close to the infrared camera are successfully differentiated in the CNN. Fig. 23(b) and (c) show the results of color-coding according to defect size. It can be seen that the distribution is clustered by size. According to Fig. 23(b) and (c), the defect size also contributes to the classification of the latent variable space.

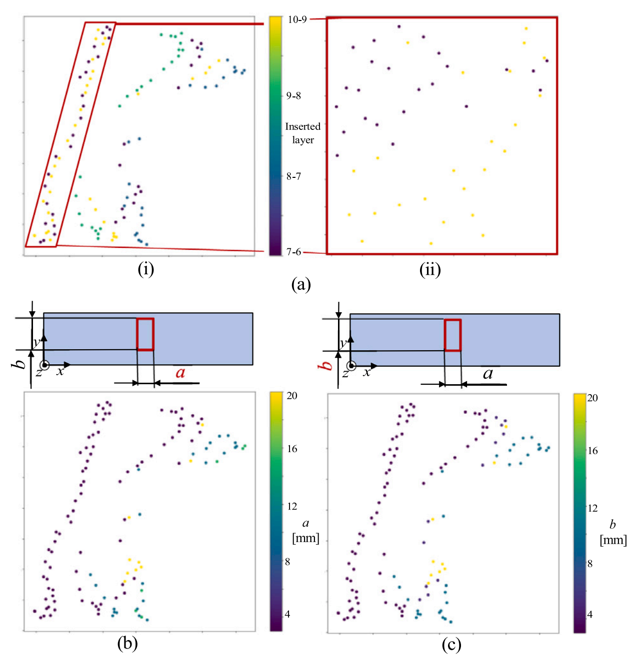

Fig. 23. Visualization of DSPSS in latent variable space. (a) Results of color-coding by defect insertion layer (i) for all data and (ii) for data between layers 10-9 and 7-6, which are classified in the neighborhood and visualized in the latent variable space, (b) results of color-coding by defect width, and (c) results of color-coding by defect height. . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.4.2. Evaluation for latent variable space of defect using VAE

Similar to the VAE using the DSPSS, the latent variable space is visualized in a two-dimensional space using UMAP as shown in Fig. 24.

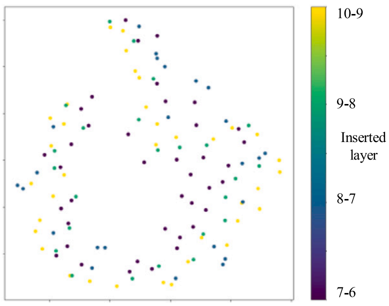

Fig. 24. Latent variable space of defect information. The 3D structure of the defect is classified by the layer number where the defect is inserted.

The figure shows that the 3D structure of the defect is classified by the layer number where the defect is inserted. From Figs. 23 and 24, it is evident that in the latent variable space, defect information and DSPSS are classified based on the defect insertion layer, width, and height of the defects. We conclude that the structure of the dataset used in this study, which allows for the classification of data based on defect characteristics, is highly suitable for predicting 3D data from 2D data using the CNN model.

4.4.3. Discussion of datasets with latent variables

By visualizing the latent variable space of the VAE using the DSPSS and defect information in Figs. 23 and 24, we found that the DSPSS and defect information can be grouped in the latent variable space according to the layer number where the defect exists. Owing to these features, the 3D information of the defect, which is 3D information, can be predicted from the two-dimensional information of the DSPSS. This enables us to classify the DSPSS when large defects are inserted between layers away from the infrared camera as shown in Fig. 27(a) and when small defects are inserted between layers close to the infrared camera as shown in Fig. 27(b).

4.5. Adaptation of the machine learning model to multiple defects

Although the CNN model used in this study does not target multiple defects, we investigated whether it is possible to predict multiple defects using the CNN model pre-trained with DSPSS obtained from FEM. The prediction results for specimens with multiple defects inserted, using the pre-trained CNN model prepared in this study, are shown in Fig. 25. The training data did not include data with multiple defects inserted. The defect information is accurately predicted when the insertion layers are the same, but the prediction accuracy for deeper inserted defects decreases when the insertion layers are changed. This is due to the fact that the normalization process is performed for each data set in this study, and stress values with multiple defects inserted in different layers are not prepared in the training data. Fig. 26 shows the prediction results for specimens with multiple defects inserted using a CNN model trained with normalization processing applied to the entire dataset, considering the magnitude of stress values during the training stage. Although the training data did not include data with multiple defects inserted, as shown in Fig. 26, the defect information is accurately predicted even when defects are inserted in different interlayers. Therefore, by performing normalization processing on the entire dataset, it is possible to detect multiple defects that do not overlap in the stacking direction.

4.6. The time required to construct the proposed method

The time required to construct the proposed method is described below. The proposed method is divided into the phases of obtaining DSPSS through FEM and infrared stress measurement and training the CNN model. In FEM, we analyzed a total of 1033 models, both with and without defects, using Abaqus CAE 2021 on an Intel Xeon Gold 6246. Each analysis took 236 s, resulting in a total time of approximately 4067 min without parallel processing. For infrared stress measurement, we measured a total of 29 specimens with and without defects. It took 350 s from loading to the start of measurement, and 20 s for lock-in processing, resulting in 370 s per measurement. Therefore, measurement of all specimens took approximately 179 min. The training of the CNN model is divided into pre-training using DSPSS obtained from FEM and transfer learning using DSPSS obtained from infrared stress measurement. The CNN model was trained using two parallel GeForce RTX 3090 32 GB GPUs. Pre-training took 15 440 s, and transfer learning took 130 s, resulting in a total training time of approximately 260 min. The prediction of defect information using the trained CNN model takes less than 1 s. The total time required for constructing the CNN model and performing the defect prediction is approximately 4506 min. For comparison, if we were to use forward FEM analysis with 100 iterations to repeatedly search for defects, the time required would be calculated as follows: 370 s for one infrared stress measurement plus 236 s per FEM analysis multiplied by 100 iterations, resulting in a total of 23 970 s, which equals approximately 400 min. Thus, if there are 10 infrared stress measurement results, the time required for defect detection using the machine learning method would be comparable to the time required for defect detection using repeated FEM analysis. If there are 20 infrared stress measurement results, the machine learning method would be approximately twice as fast as the repeated FEM analysis method This demonstrates that as the number of analyses increases, the machine learning model becomes more efficient compared to running forward FEM analyses. The efficiency gain is especially significant when dealing with a larger number of infrared stress measurement results, highlighting the advantage of using a machine learning-based approach for defect prediction in CFRP structures.

4.7. Verification of the defect inverse estimation model using latent variables through a VAE with FEM stress distribution

In this study, the VAE was used to demonstrate that DSPSS and defect information can be classified based on the defect insertion layer, width, and height of the defects, and to show that it is possible to predict 3D defects from 2D stress distribution data. Here, using the FEM stress distribution of 480 specimens with 3.0 mm × 3.0 mm defects inserted as shown in Table 5, we demonstrate that it is possible to predict defect information using latent variables. The prediction model for inverse estimation of internal defects using the latent variables obtained from DSPSS and defect information through the VAE is shown in Fig. 28. This model is an MLP model that links the latent variables dimensionally reduced by the VAE encoder for DSPSS with those dimensionally reduced by the VAE encoder for defect. The training conditions of this model are shown in Table 11. Mean squared error during the training of the MLP model and accuracy during the training of the MLP model are shown in Fig. 29. Training the model under these conditions and evaluating the F-score of the prediction results yields a histogram as shown in Fig. 30. Fig. 31(a) shows the training data and prediction results with the highest F-score, and Fig. N(b) shows the training data and prediction results with the lowest F-score. It is demonstrated that the defect insertion layer and planar position can be accurately predicted even for test data not used in training.

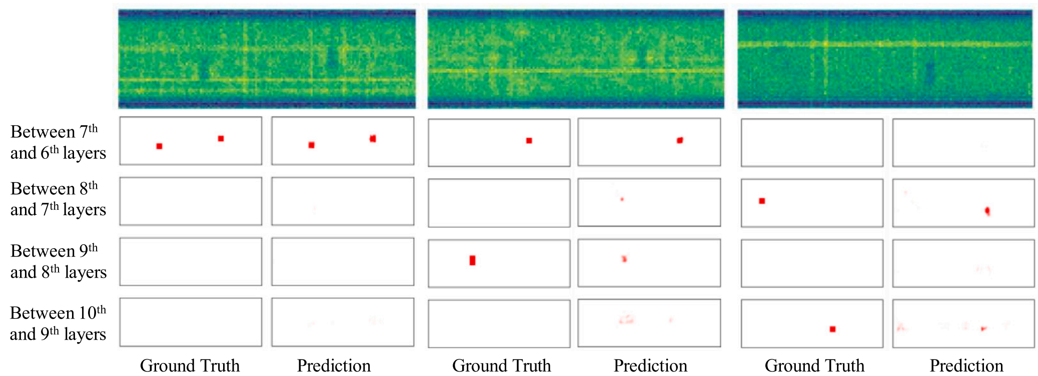

Fig. 25. Prediction results of CNN models trained by FEM-DSPSSs with per-data normalization.

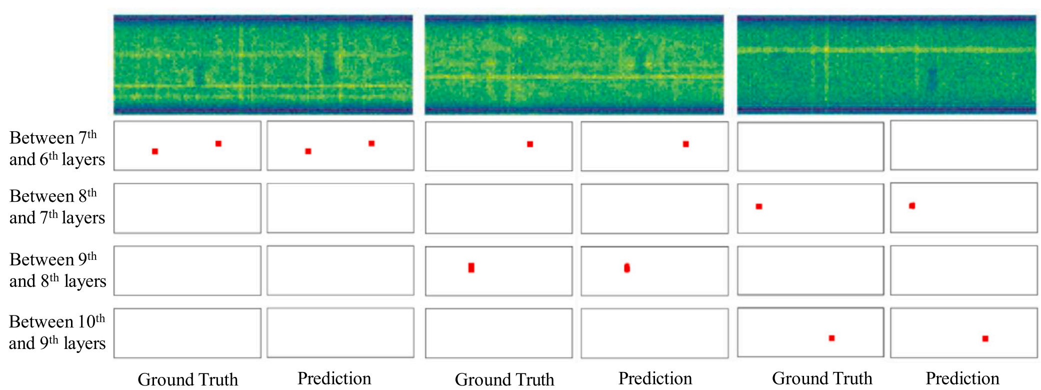

Fig. 26. Prediction results of CNN models trained by FEM-DSPSSs with normalization on the entire dataset.

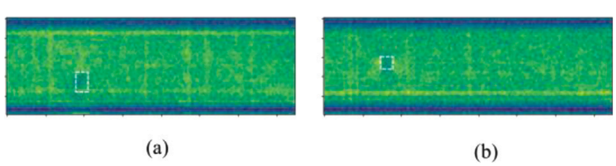

Fig. 27. (a) Stress distribution with noise obtained by FEM with 3 mm × 5 mm defect inserted between 10-9 layers and (b) stress distribution with noise obtained by FEM with 3 mm × 3 mm defect inserted between 7-6 layers. The white dashed line is the area where the defect is inserted.

Table 11
Training conditions for the defect 3D structure prediction model using latent variables.

Loss function	Mean squared error
Optimizer	Adam (Lr 0.0005)
Batch size	32
Number of training epochs	1500
Defect size [mm]	3.0 × 3.0
Number of training data	384
Number of validation data	48
Number of test data	48

5. Conclusions

A CNN predicts the 3D information of defects using the DSPSS obtained by infrared stress measurement on a simply shaped CFRP specimens with and without a defect. The CNN is pretrained with the datasets obtained by FEM. Transfer learning is used for the datasets obtained by infrared stress measurement to increase the accuracy of prediction. As a result, the following are confirmed:

- The proposed CNN can accurately discriminate the presence or absence of a defect from the DSPSS obtained by infrared stress measurement.
- The proposed CNN can predict the approximate defect insertion layer and plane position.
- The defect plane position matching ratio R is within 2 mm wider than the ground truth.
- The prediction of the defect insertion layer P in the correct layer is greater than 0.5 for all test data.
- The datasets used in this study can be clustered by the same features, which is one of the factors that made it possible to predict defect information, which is 3D data, from the DSPSS, which is two-dimensional data.

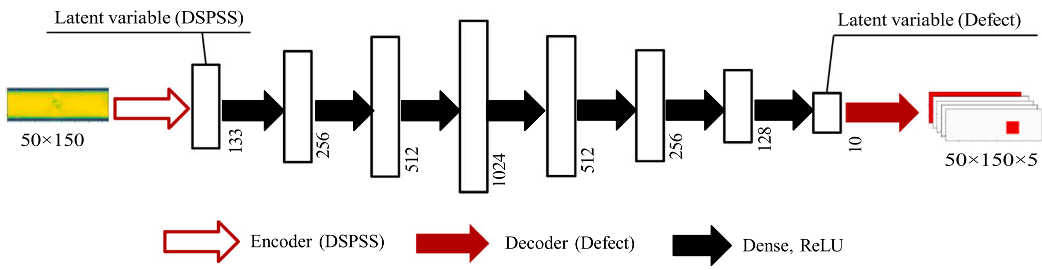

Fig. 28. Schematic diagram of the defect 3D structure prediction model using latent variables.

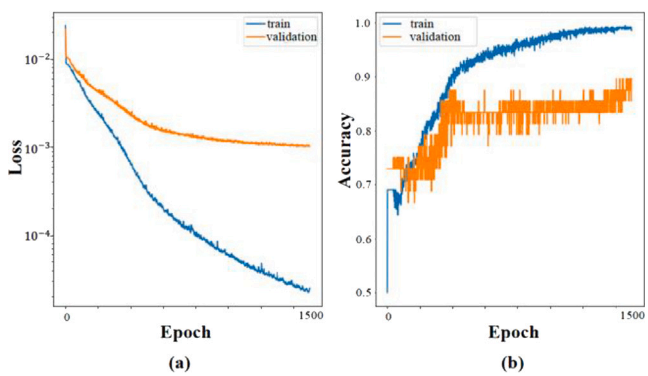

Fig. 29. (a) Mean squared loss during the training of the MLP model and (b) accuracy during the training of the MLP model.

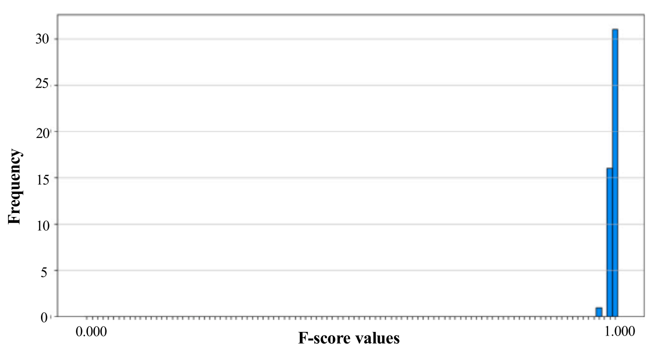

Fig. 30. The histogram of F-score in the prediction results.

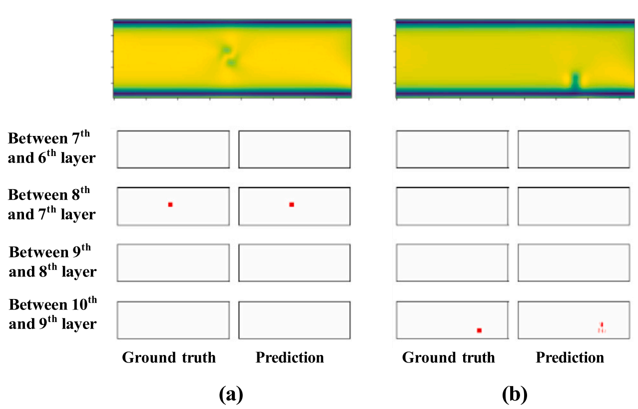

Fig. 31. Predicted results of the defect three-dimensional structure prediction model using latent variables. (a) Prediction result with the highest F-score and (b) prediction result with the lowest F-score.

CRediT authorship contribution statement

Yuta Kojima: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Kenta Hirayama:** Software, Project administration, Methodology. **Katsuhiro Endo:** Software. **Yoshihisa Harada:** Data curation. **Mayu Muramatsu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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